AI-Based Medical Diagnosis Assistant - Diagnostic Report

Automated clinical decision-support report generated from ECG, EEG or chest radiograph input.

Analysis ID	xray-20261029111424-9744476ed1
Modality	XRAY
Patient reference	PT-0003
Age / Sex	None / unknown
Referring clinician	-
Generated at	2026-10-29T11:14:24.881505Z
Model version	1.0.0
Model source	trained

Primary Diagnosis

Condition	**CONSOLIDATION**
Confidence	100.0%
Severity	Moderate
Interpretation	Dense air-space consolidation suggesting infective or inflammatory change.

Class Probability Distribution

Condition	Probability
CONSOLIDATION	100.0%
NORMAL	0.0%
PNEUMONIA	0.0%
EFFUSION	0.0%
TUBERCULOSIS	0.0%
COVID19	0.0%

Clinical Findings

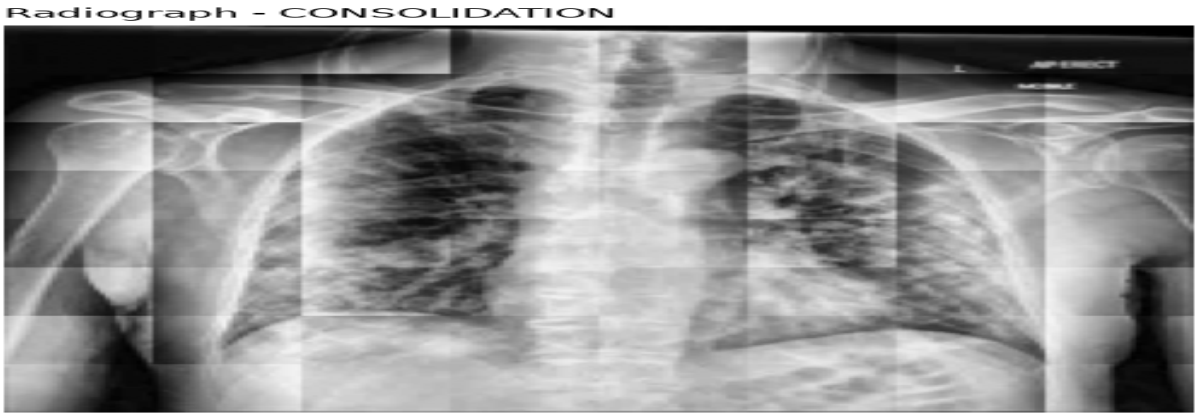
Finding	Value	Reference	Status
Mean Intensity	139.2 level	20 - 120 level	Abnormal
Lung Opacity	0.00 %	0 - 25 %	Normal
Apical Opacity	0.51 index	0 - 0.6 index	Normal
Perihilar Opacity	0.16 index	0 - 1.3 index	Normal
Perihilar Peak	0.57 index	0 - 2 index	Normal
Peripheral Infiltration	0.64 index	0 - 0.55 index	Abnormal
Costophrenic Blunting	0.57 index	0 - 0.5 index	Borderline
Symmetry Score	0.95 index	0.75 - 1 index	Normal
Texture Entropy	5.77 bits	4 - 7.2 bits	Normal
Edge Density	39.2 %	0.5 - 12 %	Abnormal
Image Quality	0.00 index	0.5 - 1 index	Abnormal

Explainable AI Analysis

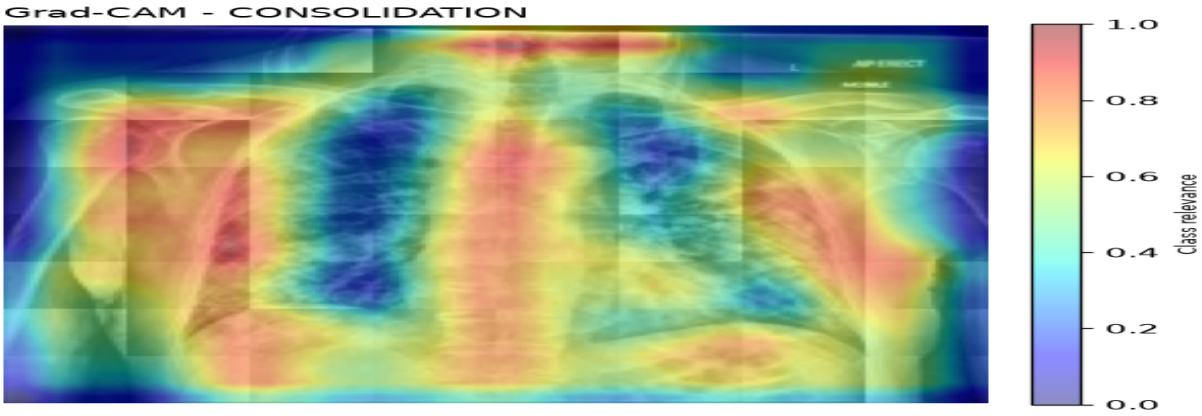
The prediction 'CONSOLIDATION' was driven mainly by Edge Density, Peripheral Infiltration, Texture Entropy. The model assigned 100.0% confidence to this class.

Attribution method: Grad-CAM over the final convolutional block

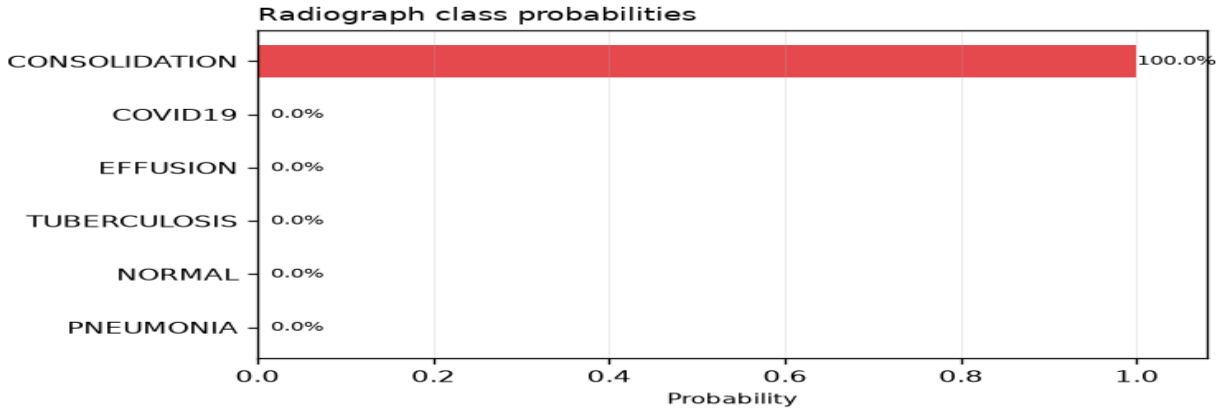
Signal attribution map



Grad-CAM localisation



Conditioned recording



Leading Feature Contributions

Feature	Contribution	Direction
Edge Density	1.000	Increases Risk
Peripheral Infiltration	0.481	Increases Risk
Texture Entropy	-0.342	Decreases Risk
Symmetry Score	-0.166	Decreases Risk
Mean Intensity	-0.128	Decreases Risk
Perihilar Opacity	0.106	Increases Risk

Perihilar Peak	0.086	Increases Risk
Costophrenic Blunting	0.062	Increases Risk

Clinical Recommendations

1. Air-space consolidation identified - evaluate for bacterial infection, aspiration or malignancy.
2. Compare with prior imaging to determine whether the finding is new.
3. A radiologist must confirm every finding reported by this decision-support tool.

Input Quality

Signal quality index: **0.0** | Usable: **no** | SNR: **0.0 dB**

This report was generated by an AI decision-support tool. It is not a medical diagnosis and must be interpreted by a qualified healthcare professional alongside the complete patient record, physical examination and confirmatory investigations.